

STATEMENT OF WORK (SOW)

Directorate of Training Sustainment (DOTS) Water Trailer Specifications for 500 Gallon Water Trailers.

Introduction: To ensure that the water trailers are available and operate properly when needed. The contractor will make sure that the water trailers meet the specifications listed below. These water trailers will be towed with allowable speeds of 80 km/h highway and 48 km/h cross-country. The water trailers we are requesting are not IAW industry standard they are IAW Department of the Army standards (higher level).

POTABLE WATER TANK (FIELD GRADE STEEL) specifications:

1. The Water Trailer must be 500 gallons oval straight barrel.
2. The Water Trailer should have no slope.
3. The Water Trailer must be made of 10-gauge Type 304L stainless steel.
4. The Water Trailers must be certified under NSF/ANSI 61 for drinking water use.
5. The Water Trailer's entire tank must be double walled.
6. The Water Trailer must have an insulated stainless-steel tank.
7. The Water Trailer must be injected with polyurethane.
8. The Water Trailer's weld must be above the water line.
9. On the Water Trailer, there will not be a crack between the tank hull and the drain that exposes the insulation.

Composition specifications:

1. The Water Trailers must have the following percent composition: Carbon (0.035 max), Manganese (2.00 max), Phosphorus (0.05 max), Sulfur (0.030 max), Silicon (1.00 max), nickel (8.00-13.0), chromium (18.0-20.0), iron (balance).
2. The Water Trailers must be certified under NSF/ANSI 61 to show that they are approved for use with drinking water and made of food grade materials. Please reference Annex C and tables 3.1 and 3.2 (Material-specific analyses).
3. The Water Trailers should not have a stainless-steel tank temperature that will exceed 212F (100C). This is important because Cyanide gases may be released when foam is heated above this temperature.

4. The Water Trailers must have an entire weld that is continuous without any gaps. This is essential because gaps cause leaks.
5. The Water Trailers heads shall be flanged style for strength made of 10-gauge T304L stainless steel.
6. The Water Trailers must have a **drain plug** at the bottom of the trailer.

WINGS, BOLSTERS AND SIDEFILLERS specifications:

1. The Water Trailer's wings, bolsters, and side fillers must be made of stainless steel.

TANK MOUNTING specifications:

1. The Water Trailer's tank shall be mounted directly to trailer decking.

WELD TANKS FINISH specifications:

1. The Interior welding will be done using W4 and/or W5 steel.
2. The welding will have a smooth finish.
3. The welding will be done using a full stitch method.
4. After the welding process is complete, no seams should be visible in the final product.
5. The drain plug in the bottom of tank will be welded using a plug weld.
6. All seams will be welded on the exterior of the tank.

MANHOLE specifications:

1. The Manhole, one (1) each, 20", T-304 stainless steel, quick opening, with a single cam latch.
2. The Manhole will have a double lid rubber seal that is comprised of 1-2-inch thickness.
3. The Manhole will have a hinge stop for keeping the lid off the shell when open.
4. The Manhole will have a locking provision for the government; contractor will supply pad locks.
5. The Manhole will be in the top center proximately and hinged at the front.
6. The Manhole Cover will have a double gasket (one on tank and one on manhole cover).

VENTING specifications:

1. The Water Trailer should have one (1) 1/2" bronze vent(s) located in the manhole lid.

TRAILER LIGHTING specifications:

1. The Trailer Lighting should be dual voltage (12 and 24 Volts) to meet D.O.T. specifications.
2. The Trailer Lighting will be wired with a NATO electrical plug. This is mandatory because primary towed vehicles will be the LMTV 2-1/2 ton and 5-ton military trucks (24 Volts).

REFLECTORS specifications:

1. The Reflectors must meet D.O.T. specifications.

ELECTRICAL specifications:

1. The electrical must have a 4-pin harness (12 Volt) which will be encased in wire loom.
2. The Ground wire will be separate from the rest of wiring harness.
3. Install grommets for harnesses and ground wire to run through separately.

ELECTRICAL PLUG specifications:

1. The NATO male electrical plug must be located at the front of trailer at the coupler.
2. A 4-pin flat harness must be located at the front of the trailer or coupler.
3. The 4-Pin flat and NATO plug should be individual connectors.

PLUMBING OUTLET specifications:**PLUMBING (FRONT OF TANK):**

1. Each Water Trailer will have one manifold. This manifold includes the following:
 - a. One 1-1/2-inch outlet connected to a bronze lockable ball valve.
 - b. The valve must connect to 1-1/2-inch stainless steel piping.
 - c. The piping splits (tees) to both sides of the trailer.
 - d. On each side there must have two 3/4 inch self-closing plain end barrel faucets and one 1-inch self-closing plain end barrel faucet.
 - e. These faucets should be located just outside each frame rail.
2. The faucets shall have a stainless-steel lockable cover at each side of the trailer.

3. The spickets need to be strapped down for stability.

OUTLET PLUMBING (REAR OF TANK):

1. The Outlet Plumbing must have one 2-inch butterfly valve connected to a bronze Camlock adapter.
2. The Outlet Plumbing must have a lockable cap with chain.
3. The Trailer must have a 7,000-pound GVWR Tandem Axle Trailer.

FRAME specifications:

1. The water trailer frame will have heavy duty steel channel throughout with a 52" drawbar and 57" x 96" deck.

DECKING specifications:

1. The entire top of the trailer will be covered with carbon steel diamond plate.

BUMPER specifications:

1. The rear bumper will be made of 5" x 3" x 1/8" carbon steel.

AXLES specifications:

1. The water trailer will have two (2) each 3,500 lb. 2-3/8" round axles.
2. The Water trailer will operate with hydraulic brakes (free backing in reverse).

WHEEL ENDS specifications:

1. The Wheel Ends will have four (4) greased bearings.
2. The Wheel Ends will have four (4) bearings; each supplied with grease lubrication.
3. The Wheel Ends will have Six (6) studs with easy lube axles.

WHEELS specifications:

1. There must be four (4) steel pieces, each measuring 16 inches by 6 inches.
2. The Wheel rims to be painted the same color as the trailer (OD Green).

TIRES specifications:

1. The Water Trailer will have four (4) tires each that are 235/75R16 Cooper Discoverer HT or similar.

BRAKE SYSTEM specifications:

1. The Brake System will have hydraulic surge brakes for tandem and an axle trailer with emergency break-away braking.

HITCH specifications:

1. The Hitch must contain one surge brake coupler with 3-inch lunette eye and a minimum of three (3) adjustable positions.
2. The Hitch will have a 12,000 pounds load capacity.
3. The Hitch height of the lunette eye needs to be set for 35 inches, and 21 inches.

DOLLY JACK specifications:

1. The Dolly Jack one (1) each will be heavy duty with 2,000 lb. capacity. Each Dolly Jack will be on heavy duty with a 2,000-pound capacity.
2. The Dolly Jack must be retractable.
3. The Dolly Jack wheel will be mounted to trailer tongue.

FENDERS specifications:

1. Each fender will have two carbon steel parts.
2. The Fenders will have a mud splash-back plate.

MUDFLAPS specifications:

1. The Mudflaps must be installed at the rear of fenders.
2. The Mudflaps must be plain black rubber.

SAFETY CHAIN specifications:

1. Each of the two safety chains (one on each side of the tank) must be 5/16 inch thick and have a breaking strength of 18,800 pounds.
2. The Safety Chains will have hooks.
3. The Safety Chains must incorporate hooks.

TIE DOWNS specifications:

1. The Water Trailer must have four (4) tie-down rings, one (1) at each corner of the trailer's exterior, are required for securing the loads.

PAINT specifications:

1. The Water Trailer will be OD Green (no gloss).
2. The Water Trailer Wheels will be painted to match the trailer's paint color.
3. The stainless-steel tank will be painted to match the trailer's paint color.

PLUMBING WILL NOT BE PAINTED.**MARKINGS specifications:**

1. The words "POTABLE WATER ONLY" are stenciled on both long sides of the containers' exterior per TBD MED 577, Chapter 8-8.
2. Two-inch letters and numbers reading "500 GALLONS" will be stenciled in black on the tank on both long sides under "POTABLE WATER ONLY".

MANUALS specifications:

1. Each individual Water Trailer will be supplied with an operation and maintenance manual shipped with the Water Trailer.
2. There will be one Operator & Maintenance Manual shipped with each water trailer.
3. There will be a wiring schematic with each water trailer.
4. There will be an equipment parts list with each water trailer.

PASSIVATION specifications:

1. The complete interior of tank must be passivated to remove the free iron from the surface of the metal using an acid solution to prevent rust.
2. The complete interior surfaces of the tank must be treated through passivation by applying an acid solution to dissolve free iron particles and promote a protective oxide layer, preventing rust.

DATA PLATE specifications:

1. There will be a government data plate with the contract information on it with the company name and the phone number.

UID specifications:

1. The Water Trailers will have a UID (Unique Identification) plate installed per MIL-STD-130N, on the trailer.

Pick-Up/Drop-Off Locations:

1. The Government POC will plan for delivery of water trailers.
2. There will be one assigned location for delivery at Fort Benning.
3. The contractor will provide their own transportation to come to Fort Benning to deliver the water trailers.

Inspection: Pick-Up/Drop-Off Locations:

1. There will be a 100% inspection upon delivery of the water trailers between both parties.
2. The Water Trailers must pass food grade inspection with no possibility of rust or seepage into seam.
3. An Internal Final Inspection and a leak proof test of all sealed areas shall be performed prior to, and in preparation of a Customer/Government Inspection.
4. If any deficiencies are found, they shall be corrected before unit is shipped or corrected before payment.
5. Representative of the Contracting Provider will be present for testing and serviceability prior to acceptance at the receiving location.

Maintenance/Repair:

1. The Unit shall be guaranteed to be free from defects in material and workmanship, while under the normal use and service by the original purchaser, for a period of ONE (1) year from the date unit is delivered.
2. The local repair facility will be sponsored.

Emergency Service/Repair:

1. Due to the critical nature of availability and mission readiness, the vendor will come to Fort Benning as needed to perform required services on water trailers if they are rendered non-mission capable upon delivery and cannot be replaced in a timely manner.
2. The vendor will make onsite repairs if capable.

Testing:

1. An Internal Final Inspection and a leak proof test of all sealed areas shall be performed prior to, and in preparation of a Customer/Government Inspection.
2. If any deficiencies are found, they shall be corrected before unit is shipped.
3. Representative of the Contracting Provider will be present for testing and serviceability prior to acceptance at the receiving location.

ROAD TEST:

1. The Water Trailer will be road tested.
2. The Water Trailer will be pulled behind a vehicle at highway speed empty and full to check for sway.
3. The Water Trailer will be pulled behind a vehicle, and a break stop test will be performed.
4. All light functions will be checked including both 12 and 24-volt.
5. NOTE: All brand name parts mentioned in specifications above are subjected to substitution of equal part that is of a different brand.

To be considered for award, offerors must strictly adhere to Government and shall not substitute for or interpret them based on private industry standards.